

Nestora - AI Bathroom Studio

AI-assisted bathroom planning with deterministic spatial validation

The problem

Bathroom planning combines space constraints, fixture placement, aesthetics, product selection and budget - decisions that are difficult to evaluate together.

The solution

Nestora turns a user brief into validated layout options, an interactive 2D/3D design, a budget-aware product bundle and an explainable AI review.

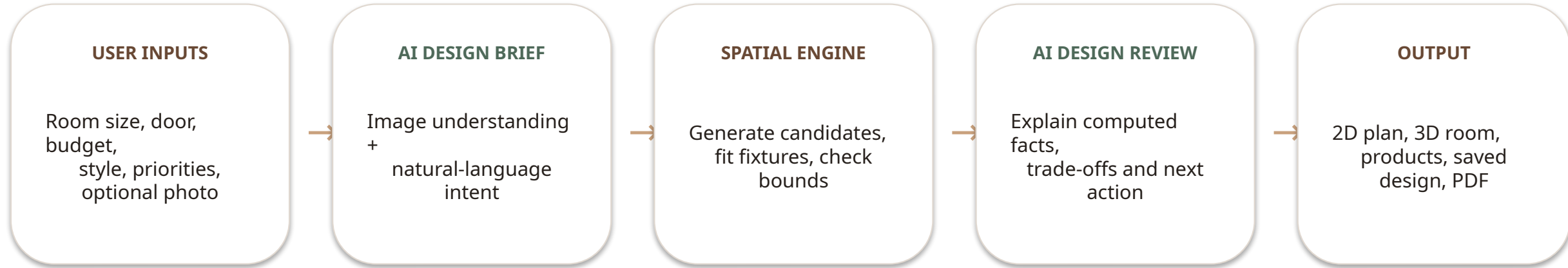
One planning flow

- 1 Room + priorities
- 2 **Layout generation**
- 3 **Spatial validation**
- 4 2D + 3D review
- 5 Products + AI review

AI reasons. The spatial engine validates.

How Nestora Works

A hybrid architecture separates AI interpretation from measurable geometry



Why hybrid?

Gemini interprets images, user intent and design evidence. Deterministic code remains responsible for fixture footprints, room bounds, layout validation and budget logic.

This avoids treating fluent AI output as proof that a physical layout fits.

Working Prototype & Technology

The prototype connects planning, visualization and AI in one web application

AI

Gemini 2.5 Flash

Image analysis
Design intelligence
Natural-language refinement

BACKEND

FastAPI + Python

Planning APIs
Spatial logic
Saved designs

DATA

SQLite + SQLAlchemy

Product catalog
Design persistence
Budget data

VISUALIZATION

SVG + Three.js

Interactive 2D plan
3D bathroom view
Shared layout state

Prototype capabilities

- Multiple strategy-based bathroom layouts with spatial checks
- Interactive 2D editing synchronized with the 3D scene
- Budget-aware fixture and product selection
- AI image analysis, design review and natural-language refinement
- Saved designs and downloadable design report

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What Makes Nestora Different

Explainable AI around a deterministic planning core

- 01 Spatially grounded** AI commentary is based on computed layout facts rather than invented measurements.
- 02 Design alternatives** Users can compare layouts built around circulation, storage and balanced-space strategies.
- 03 Budget aware** Core fixtures, optional products and upgrades are selected against the target budget.
- 04 Easy to review** The same design state drives the 2D plan, 3D view, saved design and report.

Nestora combines AI-assisted design reasoning with measurable spatial validation to make bathroom planning more understandable, interactive and practical.